Tableau Assignment Theory Question

Part A — Connect & Understand the Data

Q: List 3 fields you would classify as Dimensions and 3 as Measures from this dataset.
Ans: Dimensions (categories/labels):

- Market — Central, East, South, West
- Product Category — Tea, Coffee, Bakery, Specialty Coffee, Espresso
- Customer Type — Retail, Wholesale

Measures (numbers):

- Sales
- Profit
- Quantity

Dimensions describe or group the data. They answer questions like what, who, or where. Measures are numerical values that can be added, averaged, or compared to understand the data better.

Q: Name one field you would call discrete and one you would call continuous, and explain why in one sentence.

Ans:

1. Product Category is **discrete** because it contains separate categories such as Tea, Coffee, Bakery, Specialty Coffee, and Espresso.

2. Sales is **continuous** because it is a numerical value that can have many different values, such as $12.40, $317.65, or $1,204.00.

Q: State whether you used a Live connection or an Extract for this exercise, and why that is the right choice here.
Ans: I used an Extract because the BrewPoint dataset is a static Excel file and does not change while creating the dashboard. An extract stores a copy of the data in Tableau, which helps the charts, filters, and calculations work faster. A Live connection would be more useful if the data were coming from a database that was being updated regularly.

Part C — Intermediate Charts

Q: Build a scatter plot of Sales vs. Profit, colored by Product Category, and note any outliers you see.

Ans: The clearest outlier is Bakery. It is the only category with an overall loss, with about −$1,408 profit on $14,144 in sales.

On the other hand, Specialty Coffee is one of the strongest categories, with around $14,820 profit on $57,186 in sales.

Looking at individual orders, 212 out of 1,400 orders have negative profit. These points appear below the zero-profit line in the scatter plot and are mainly seen in Bakery and some Wholesale orders with high discounts.

Part D — Calculated Fields

Q: Create a second calculated field using IF logic that labels each order line "High Margin" or "Low Margin" based on a Profit Ratio threshold you choose. State the threshold you picked and why.
Ans:
I chose 20% (0.20) as the threshold.

The Profit Ratio is calculated as Profit ÷ Sales. In this dataset, the median Profit Ratio is close to 20%, so using 20% as the cutoff gives a simple way to divide the orders into High Margin and Low Margin groups.

Part E — Filters

Q: Add a context filter on one field of your choice, and write one sentence on why you chose that field for context rather than a regular filter.
Ans: I used Customer Type (Retail vs. Wholesale) as the context filter because Retail and Wholesale orders have different quantities and pricing patterns. Using it as a context filter helps analyze each customer type separately before applying other filters.

Part F — Capstone Dashboard

The final dashboard combines the Sales by Category bar chart, Sales Trend line chart, and Profit highlight table (Market × Product Category).

I also added a filter action so that when a user clicks a category in the Sales by Category chart, the other charts automatically show data for that category. This makes it easier to explore the dashboard without manually changing filters.

The same Product Category colors were used across the charts so that the dashboard looks consistent and is easier to understand.

Dataset Reference

The BrewPoint dataset contains:

- 1,400 order lines
- January 2024 – December 2025
- 4 markets
- 5 product categories
- 2 customer types